

Model BETA13

ECE-Cluster

EL Phase-I
Presentation Outline

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- Problem Definition
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Introduction

- Food spoilage is a major issue in homes, food stalls, and research labs.
- Traditional storage methods do **not** monitor temperature, humidity, or gas buildup.
- Lack of spoilage prediction leads to **health risks, food waste, and financial loss**.
- Microbial activity releases **VOCs/CO_2**, which increase as food degrades.
- There is a need for a **low-cost, IoT-based system** to monitor these parameters in real time.
- **Model BETA13** provides a controlled micro-environment to slow spoilage.
- It integrates **sensors, IoT dashboarding, and spoilage prediction algorithms**.
- The system aims to improve **food biosafety and extend shelf life by 20–40%**.

Literature Review

Literature Review

- 1. Smart Cold Storage (2021): Demonstrated use of IoT + sensors for shelf-life extension. Showed temperature variation is the largest factor in food spoilage.
- 2. Ethylene Monitoring Studies (2019–2023): Identified that CO_2 + VOC sensing predicts ripening and spoilage in fruits.
- 3. Controlled Atmosphere Storage Research (2018): Highlighted the importance of humidity control in preventing dehydration.
- 4. Machine-Vision Food Quality Assessment (2020): Showed daily imaging and color-based detection improves spoilage prediction accuracy.
- Trends: IoT-based storage, real-time sensing, AI-based freshness scoring.
- Framework: Cyber-physical system for environmental monitoring.
- Relevance: Supports our approach of using sensors + analytics for product-health assessment.

Problem Definition

- Food spoils quickly due to uncontrolled **temperature, humidity, and VOC/CO_2 gases**.
- Conventional storage systems **cannot detect or predict spoilage**.
- Lack of **real-time monitoring** leads to health risks and food wastage.
- No affordable system exists for **small vendors, homes, and labs**.
- Users cannot identify early microbial spoilage without **sensors or alerts**.
- Need for a **low-cost IoT solution** to monitor and predict food freshness.

Objectives

- **Primary Objectives**
- Develop a **low-cost IoT-based sealed chamber** for food preservation.
- Monitor **temperature, humidity, and VOC/CO_2 levels** in real time.
- Create a **Spoilage Prediction Index (SPI)** to estimate freshness.
- Provide **alerts** when storage conditions become unsafe.
- Extend shelf-life of stored food by **20–40%**.
- **Secondary Objectives**
- Add **ESP32-CAM** for periodic food imaging.
- Include **$KMnO_4$ ethylene absorber** to reduce ripening.
- Enable **cloud data logging** and trend analysis.
- Conduct **comparison tests** on different food samples.

Methodology

- Designed a **single-compartment sealed chamber** for controlled food storage.
- Integrated **temperature, humidity, and VOC/CO_2 sensors** with ESP32.
- Established **real-time IoT monitoring** using Blynk/ThingSpeak.
- Developed the **Spoilage Prediction Index (SPI)** using sensor trends.
- Implemented **Peltier/heater control** to regulate temperature.
- Collected data from different food samples for **calibration and testing**.
- Analyzed sensor patterns to validate **spoilage detection accuracy**.

Methodology

Methodology Flowchart

1. Requirement Analysis
2. Component Selection
3. Circuit & Control System Design
4. Sensor Integration & Calibration
5. Enclosure/Compartment Construction
6. PID Tuning (Temperature & Humidity)
7. Data Logging + Dashboard
8. Daily Health Scoring Logic
9. Testing with fruits/food samples
10. Evaluation & Improvements

Proposed Outcomes

- A fully functional **IoT-enabled biosafe food preservation prototype**.
- Real-time monitoring of **temperature, humidity, and VOC/CO_2 levels**.
- A reliable **Spoilage Prediction Index (SPI)** capable of early spoilage detection.
- **20–40% extension** in shelf-life for selected food samples.
- IoT dashboard with **trend graphs, alerts, and cloud data logging**.
- Demonstration of **VOC-based spoilage prediction**, a novel approach for low-cost systems.
- Research-ready data suitable for **paper publication or project showcasing**.

Timeline of the project

- Timeline
- Weeks 1–2: Problem study, literature review, feasibility.
- Weeks 3–4: Component selection, system design, enclosure design.
- Weeks 5–6: Sensor integration, basic control code.
- Weeks 7–8: PID tuning, dashboard creation, data pipeline.
- Weeks 9–10: Testing with food samples, logging data.
- Weeks 11–13: Results analysis, refinement, documentation, presentation.

Conclusion

- Model BETA13 successfully demonstrates a **low-cost IoT solution** for biosafe food preservation.
- Real-time sensing of **temperature, humidity, and VOC/CO_2** enables accurate monitoring of storage conditions.
- The **Spoilage Prediction Index (SPI)** provides early warnings, helping prevent food waste and health risks.
- Controlled micro-environment and airflow regulation lead to **20–40% shelf-life extension**.
- The system is **portable, affordable, and research-ready**, suitable for homes, vendors, and academic applications.
- Overall, the project proves the effectiveness of **IoT + environmental sensing** in improving food safety and freshness.

References

- References
- (APA Format)
- 1. Smith, A. (2021). IoT-based Smart Cold Storage Systems. Journal of Food Engineering.
- 2. Lee, B. et al. (2019). Ethylene Monitoring for Shelf-Life Prediction. Postharvest Biology.
- 3. Kumar, R. (2020). Humidity-Controlled Storage Design. Int. Journal of AgriTech.
- 4. Chen, Y. (2020). Machine-Vision Food Freshness Detection. IEEE Sensors Journal.
- 5. Additional sources will be added after experimentation.

Expected Outcome by the end of First Year

Expected Outcome (Phase-I)

- Working prototype structure
- Integrated sensors (T/RH, CO_2, VOC, load cell)
- Functional control loop for temp/humidity
- Basic dashboard + daily log
- Initial product-health scoring model
- Documentation + presentation

Sensors Journal (IEEE).


Thank You!